



MICROECONOMICS

PRODUCTION FUNCTION

Chapter 5 | Lecture 2

Short Run Production Function

Production ↑

1 Factor
change

Variable Factors

- Labour → change ↑
- Raw material → Same
- Fuel → Same
- Power → Same

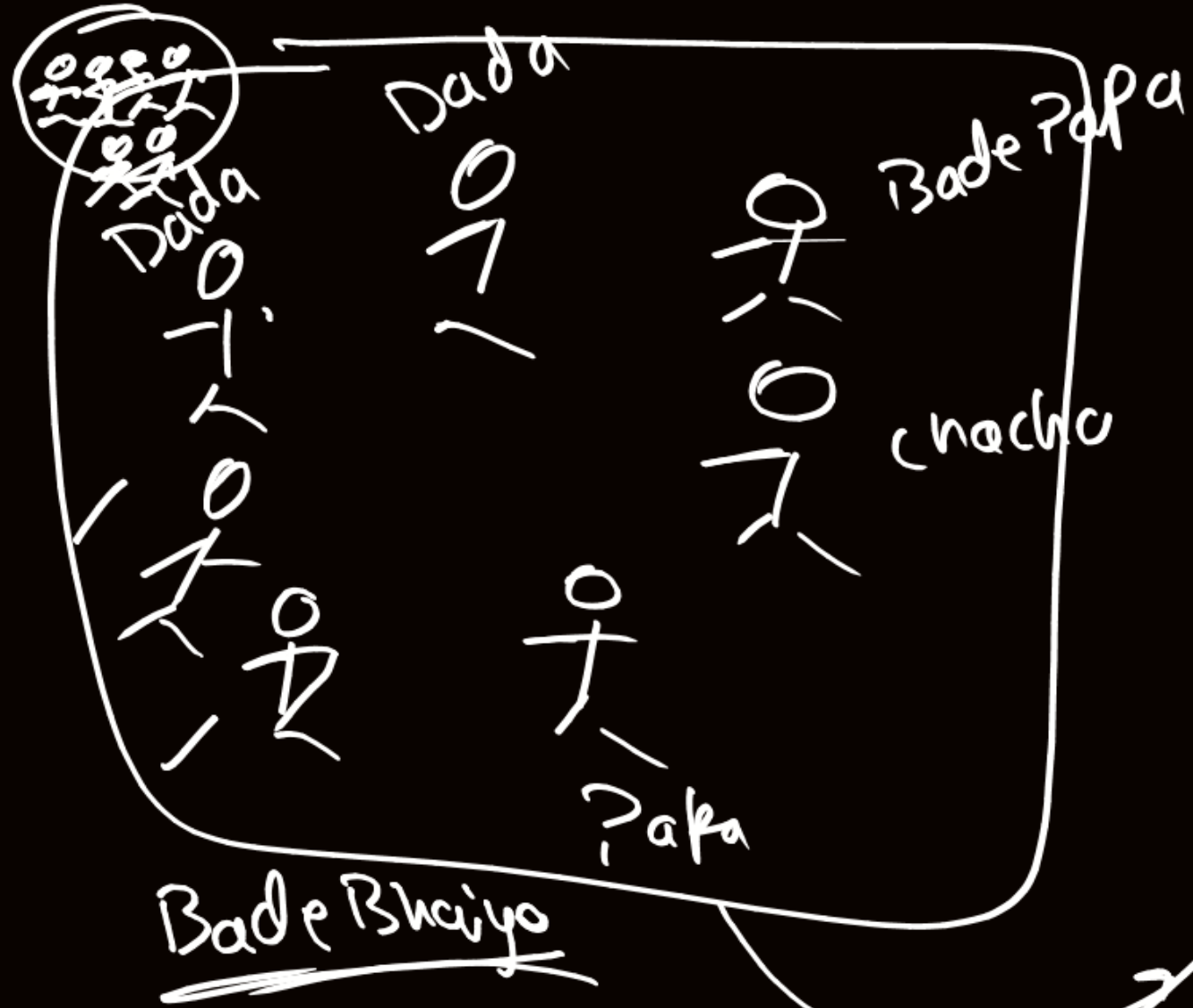
Long Run Production Function

Production ↑

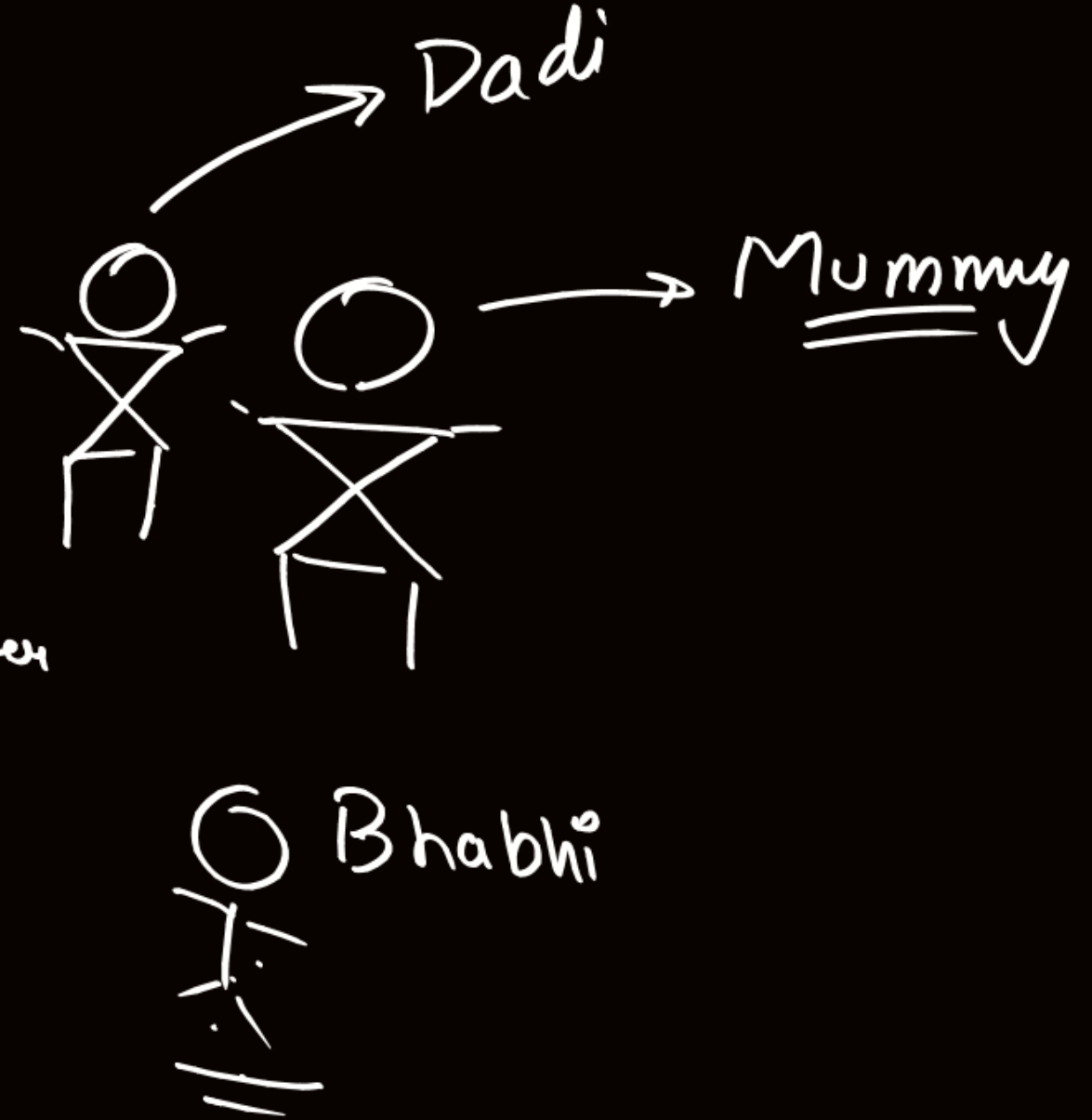
Change

- land → 50%
- labour → 50%
- Fuel → 50%
- Machine → 50%
- Raw material → 50%

LVP



Choti Sister



LUP



Law

Law of Variables Proportion (LVP) states that as we increase quantity of only one input keeping other inputs fixed, total product (TP) initially increases at an increasing rate then at a decreasing rate and finally at a negative rate.

TP

Bitter Chinki Bitter

pinku Chinki Bitter

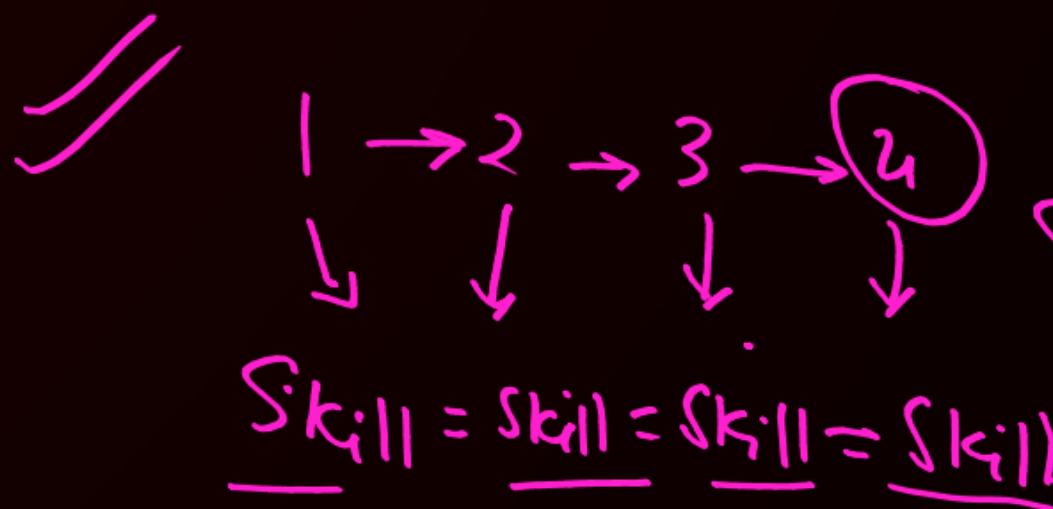


Motetisa



The assumptions of the Law of Variable Proportions in brief are:

- Fixed Technology: Technology remains constant during production.
- Variable Input: Only one input is variable, while others remain fixed.
- Short Run: The law operates in the short run where at least one factor is fixed.
- ✓ Homogeneous Input: All units of the variable factor are identical.
- ✓ Divisible Inputs: Inputs can be divided into smaller units for adjustment.



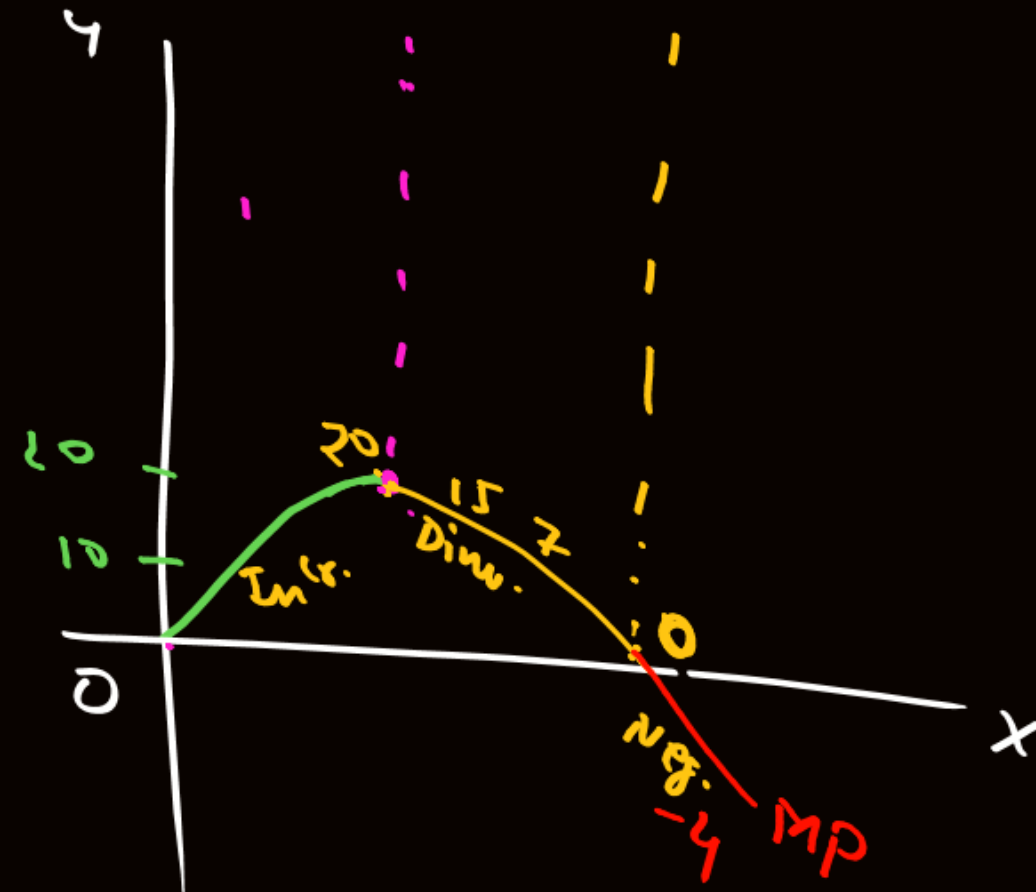
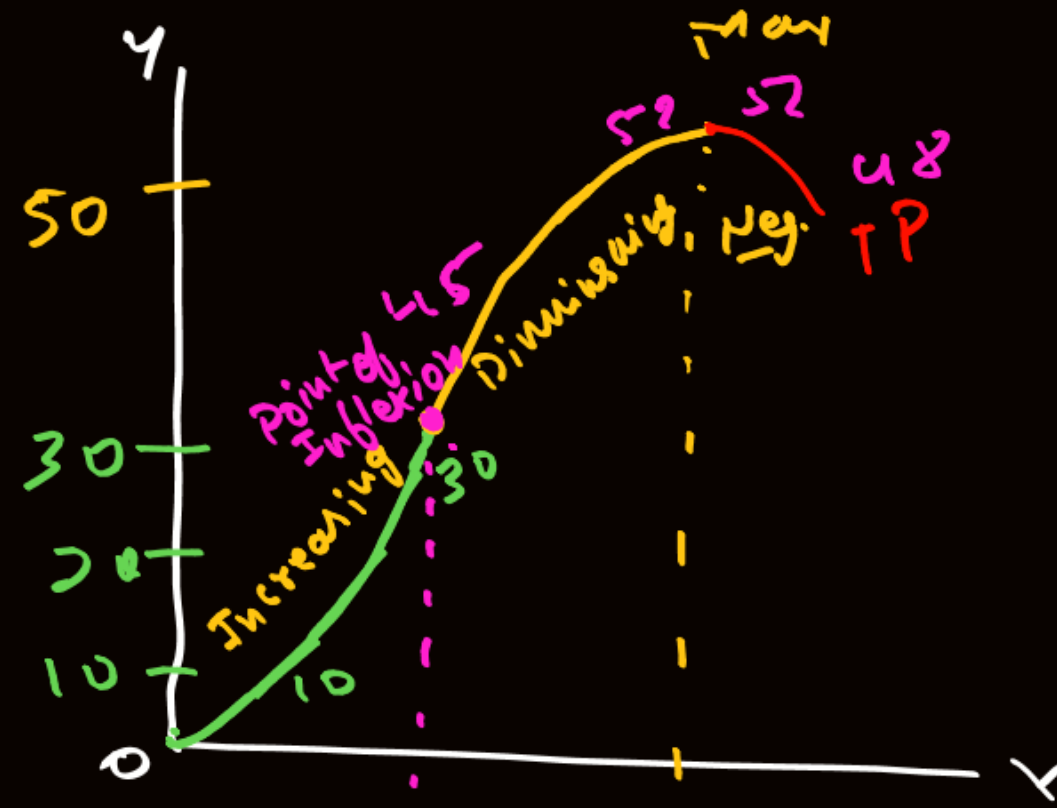
Law of Variables Proportions



① Labour

Fixed Factor (Land in acres)	Variable Factor (Labour)	TP (Units)	MP (Units)	Phase
1	1	10	10	Phase I: Increasing Returns to a Factor
1	2	30	20	
1	3	45	15	Phase II: Diminishing Returns to a Factor
1	4	52	7	
1	5	52	0	
1	6	48	-4	Phase III: Negative Returns to a Factor

Q5 = Point of Inflection
 MP = Maximum



Q.1

TP = Max
 MP = 0

Q.2

MP = Negative
 TP = Falling

Q.3

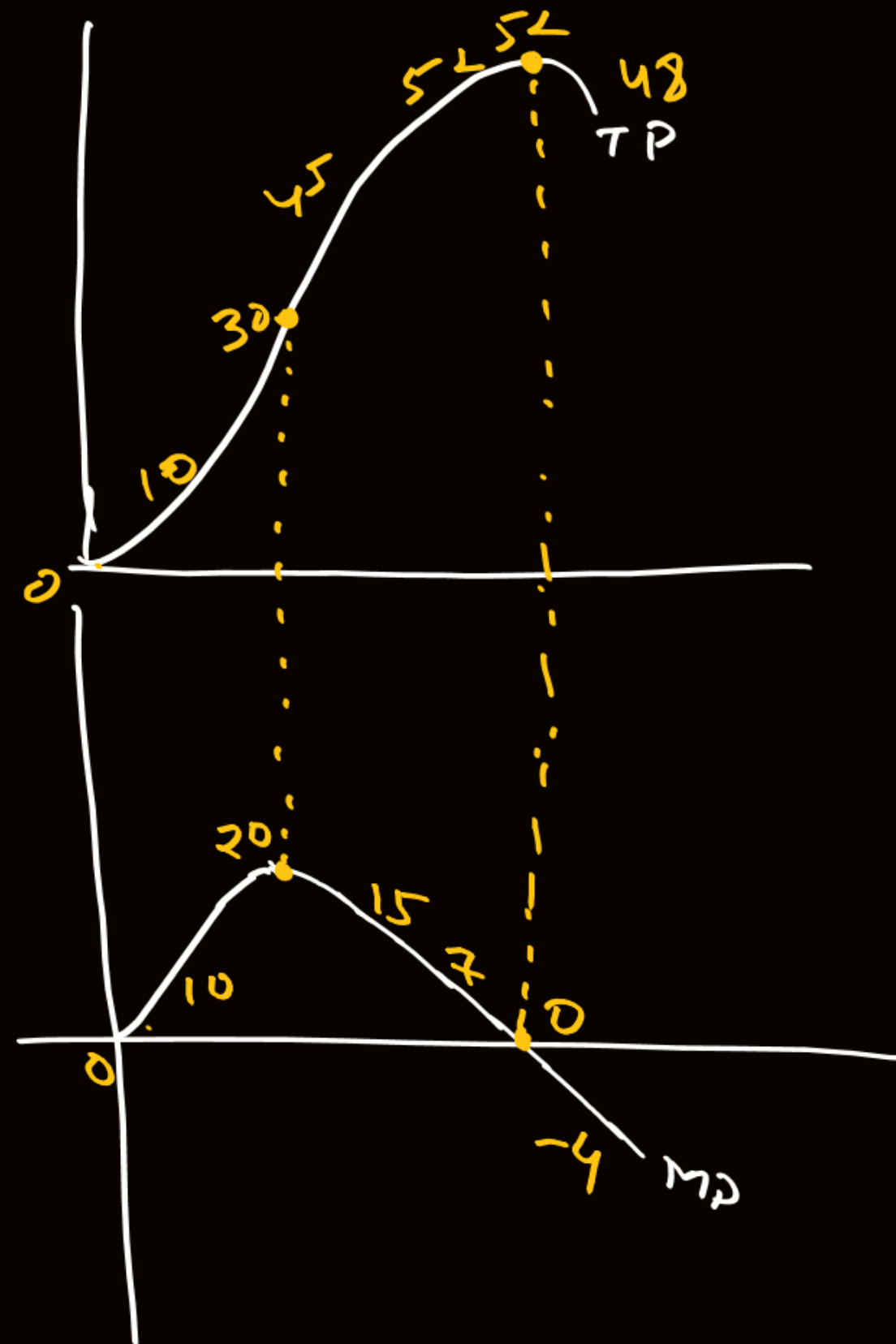
TP = rising at Increasing Rate.

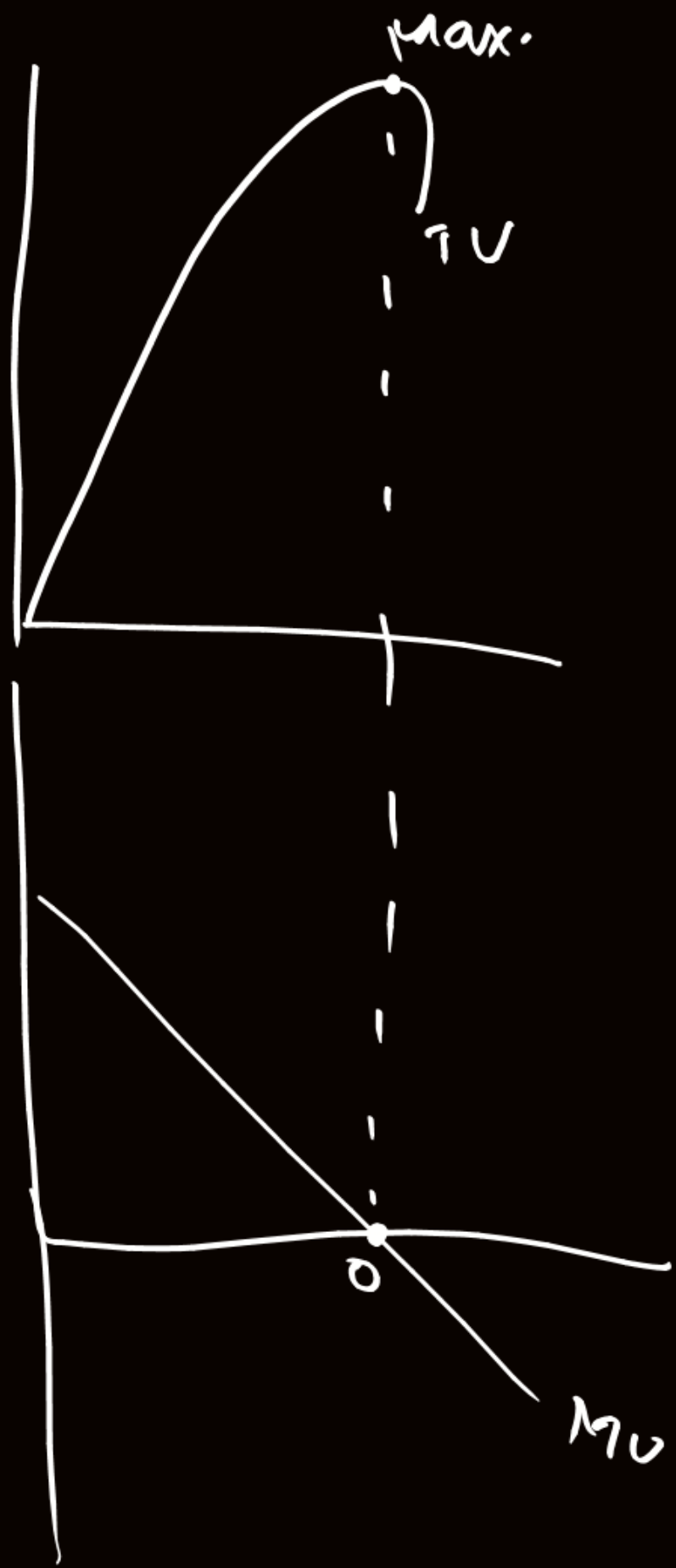
MP = Increasing

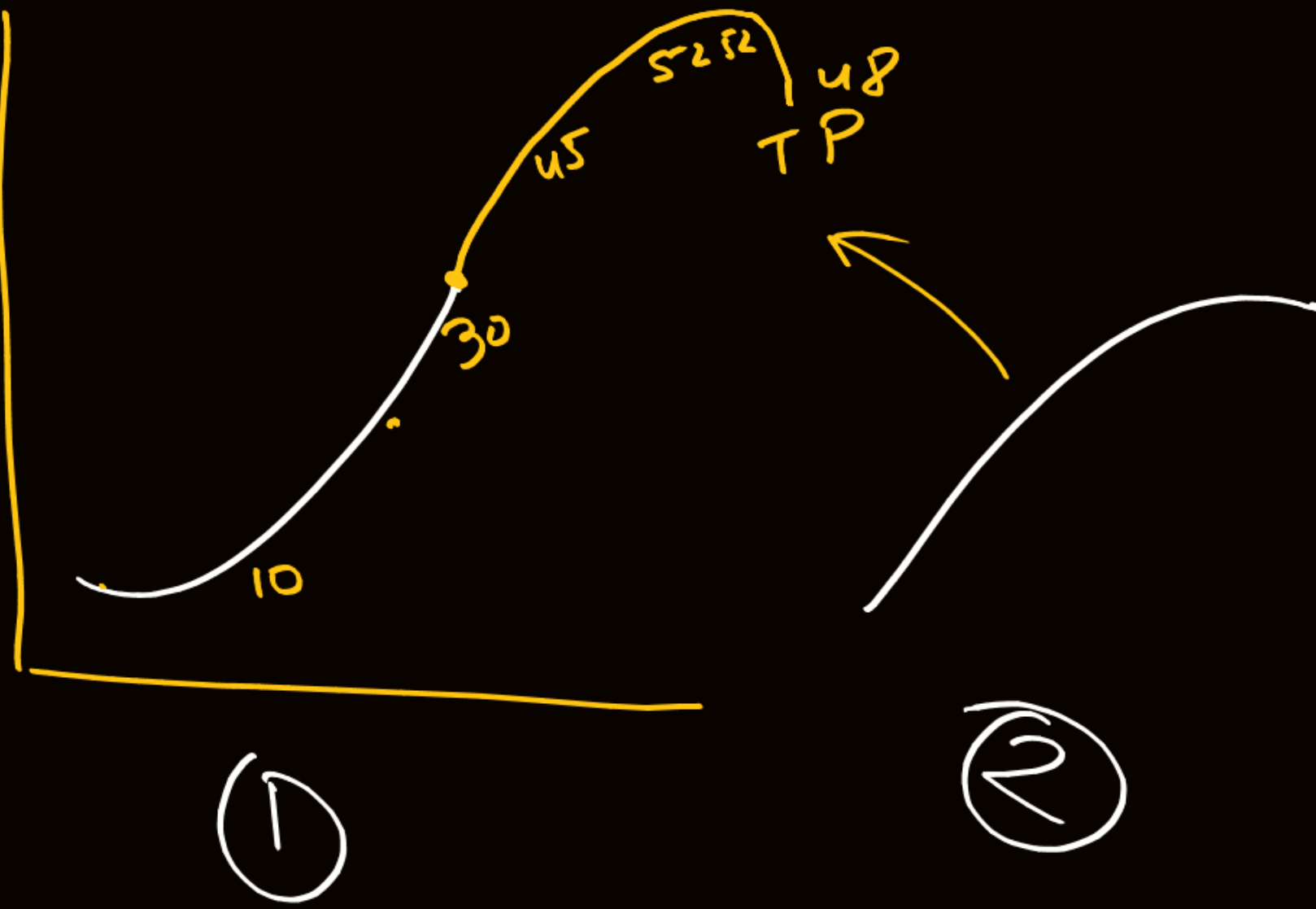
Q.4

MP = Decreasing but Positive

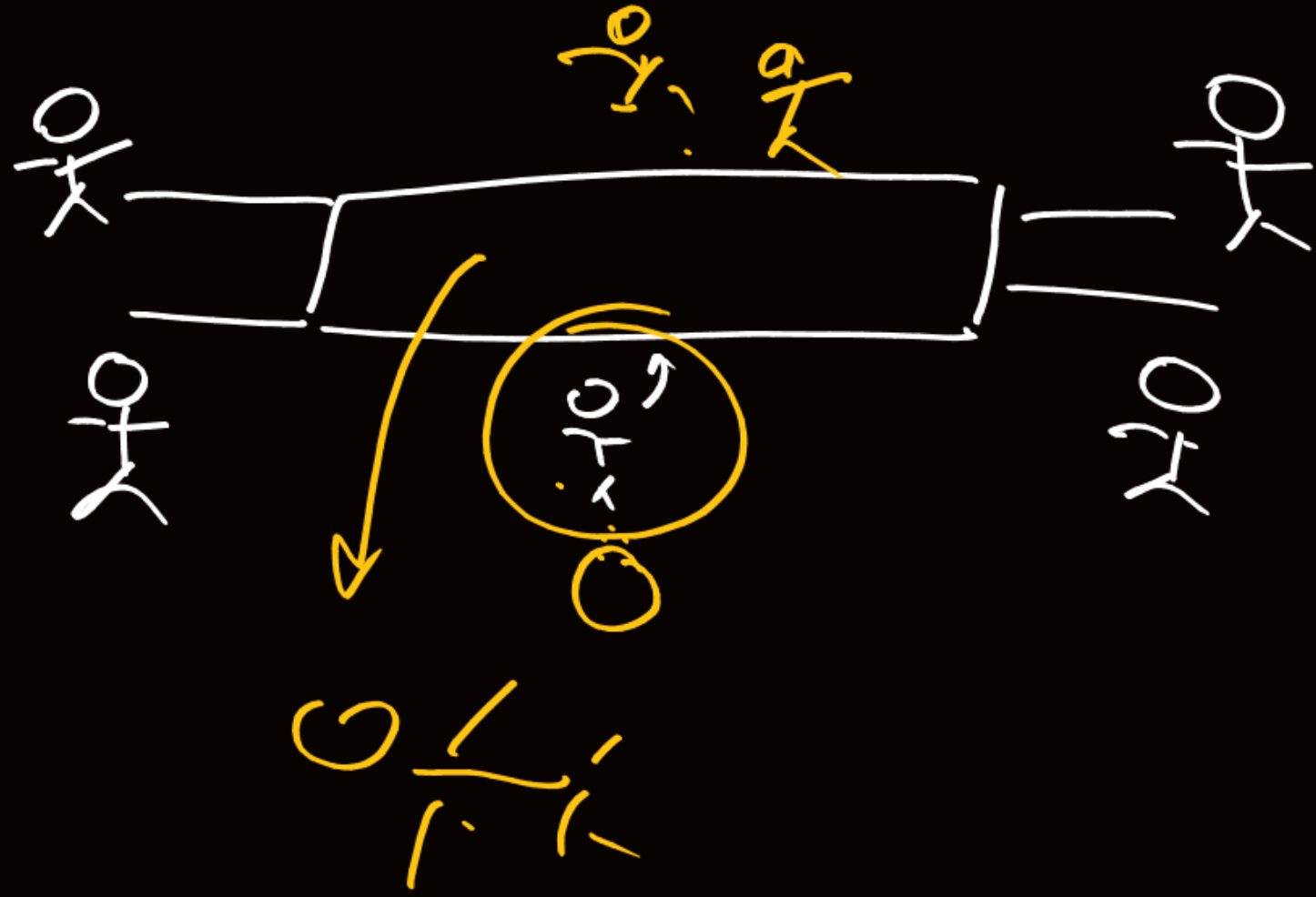
TP = Increasing at Diminishing Rate.

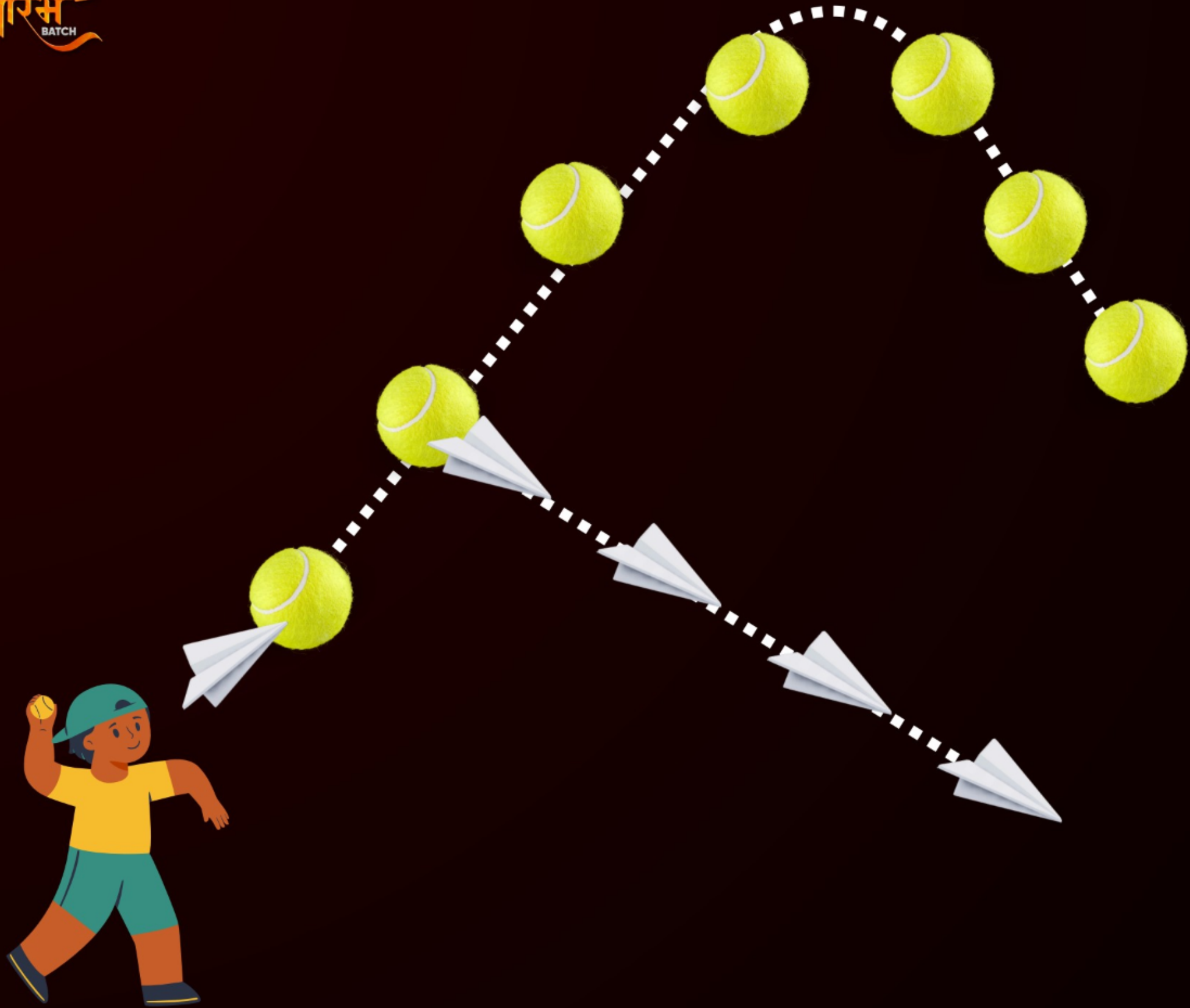


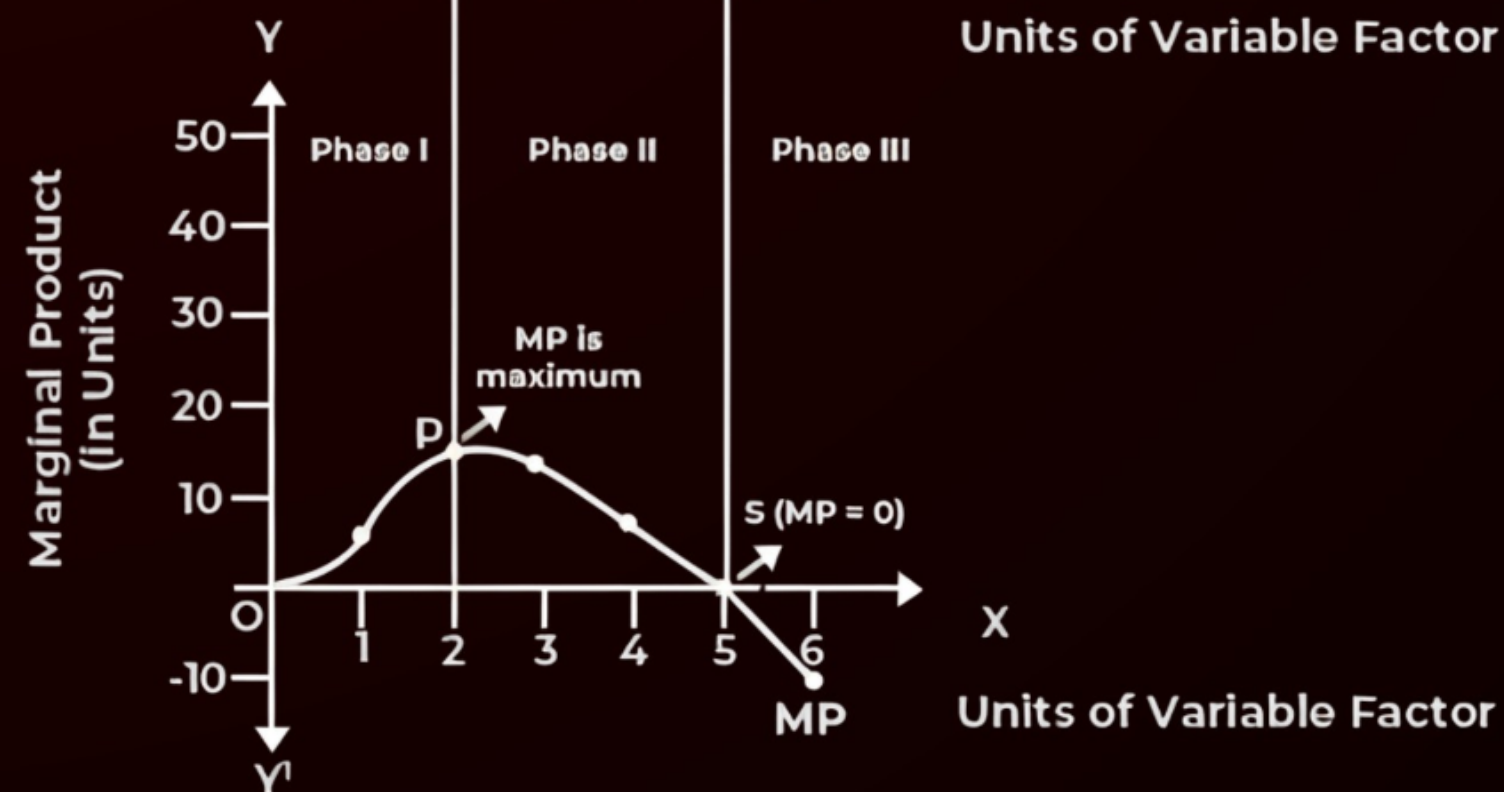
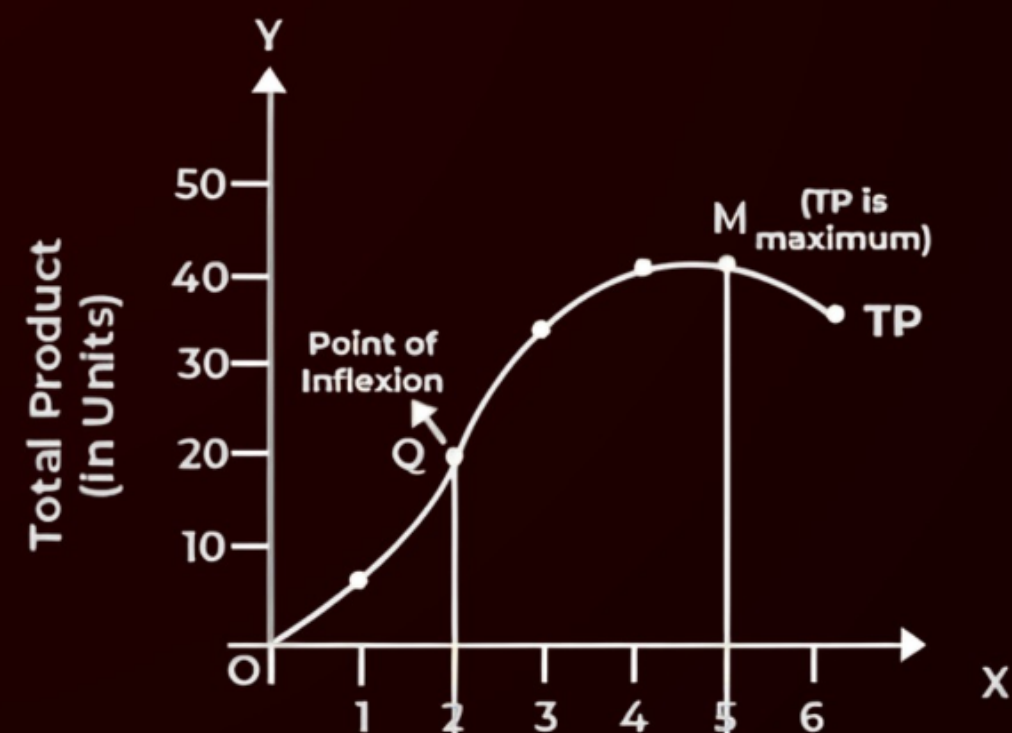




Increasing Rate







Leak

Jump

- Phase I (Between O to Q) TP increases at an increasing rate and MP also increases.
- Phase II (Between Q to M) TP increases at decreasing rate and MP falls. This phase ends when MP becomes zero and TP reaches its maximum point.
- Phase III (Beyond point M) TP starts decreasing and MP not only falls, but also becomes negative.
- Point of Inflexion (Point Q) Point 'Q' is known as point of inflexion as curvature of TP curve changes at this point.

✓ ✓

**THANK
YOU**

Stay Connected

KEEP LEARNING